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Systematic Literature Review

Measurement Properties of the EQ-5D-Y: A Systematic Review

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ABSTRACT

Objectives: This study aimed to perform a systematic review of published evidence on the psychometric properties of 3-level version of EQ-5D-Y and 5-level version of EQ-5D-Y (EQ-5D-Y-5L).

Methods: A literature search on the MEDLINE, Embase, and EuroQol website (until June 2021) was conducted. Original studies on EQ-5D-Y psychometric properties such as feasibility, distribution properties (ceiling and floor effects), reliability (test-retest, interrater, intermodal), validity (known-groups, convergent), and responsiveness, published as full-text articles in English, were included. Studies on experimental EQ-5D-Y versions were excluded. The following data were pooled using random effects models: missing values, the ceiling effect, and correlations coefficients with other measures.

Results: A total of 47 studies (inclusive of 7 on EQ-5D-Y-5L) containing data from 45 310 children and 2690 proxy respondents representing 15 countries were included. These studies were characterized as being high quality according to the quality index. The most represented areas were school populations and musculoskeletal diseases and orthopedics. The EQ-5D-Y dimensions, EQ visual analog scale, and EQ index were reported in 89%, 77%, and 26% of studies, respectively. Most articles addressed validity (known-groups, $n = 27$; convergent, $n = 21$) and reliability (test-retest and interrater, $n = 10$ each). Convergent validity studies showed that, where the assessment of the child's functioning at school is required, EQ-5D-Y should be supplemented with other school-specific measures.

Conclusions: This systematic review provides a summary of measurement properties and the psychometric performance of 3-level version of EQ-5D-Y and EQ-5D-Y-5L. The existing evidence supports using the EQ-5D-Y descriptive system and EQ visual analog scale in children and adolescent populations. Further research on test-retest reliability and the responsiveness of the EQ-5D-Y index obtained with child-specific value sets is needed.

Keywords: adolescent, child, psychometrics, quality of life, surveys and questionnaires, systematic review.

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Introduction

The EQ-5D-Y is a generic, self-complete health status instrument, which was specifically designed to measure the health-related quality of life (HRQoL) of children and adolescents. It was developed in 2010 by a group of researchers from 7 countries.¹ The EQ-5D-Y design is based on the standard adult HRQoL measure—the 3-level version of EQ-5D questionnaire. The wording of dimensions, a description of the visual analog scale (EQ VAS), and the questionnaire layout were adapted to a child-friendly format to accommodate children's specific developmental and cognitive abilities.²

The EQ-5D-Y descriptive system consists of 5 dimensions: “walking about” (the equivalent of “mobility” in the adult version of the questionnaire), “looking after myself” (“self-care”), “doing usual activities” (“usual activities”), “having pain or discomfort” (“pain or discomfort”), and “feeling worried, sad, or unhappy” (“anxiety or depression”). In the original version (3-level version of EQ-5D-Y [EQ-5D-Y-3L]), each dimension has 3 levels: “no

problems,” “some problems,” and “a lot of problems.”¹ Recently, a new version of the EQ-5D-Y (5-level version of EQ-5D [EQ-5D-5L]) was developed, with 2 additional levels to improve its measurement properties.³ The whole instrument refers to “your health TODAY.” Despite the descriptive system, the child rates his or her overall health on the EQ VAS, a vertical scale ranging from 0 (labeled “The worst health you can imagine”) to 100 (labeled “The best health you can imagine”).⁴

The EQ-5D-Y-3L descriptive system allows 243 different health states to be generated, which can be described by a 5-digit code, defined by a combination of level indicators for each of the 5 dimensions (eg, 21 123). By applying a formula that attaches values to each of the levels in each dimension, EQ-5D-Y health states may be assigned a summary index corresponding to a health state utility value (EQ index). The official EQ-5D-Y valuation protocol (the methodology to obtain the formula) was published in July 2020.⁵ Value sets developed with this protocol should enable EQ-5D-Y to be used in cost-utility analyses of medical interventions in children.^{6,7}

The development of the EQ-5D-Y-3L questionnaire was originally supported by the multicountry study on its measurement properties.⁸ Ravens-Sieberer et al⁸ provided preliminary evidence of the instrument's feasibility, reliability, and validity. Nowadays, the complete article and tablet versions of EQ-5D-Y-3L are available in >90 and 70 languages, respectively. Between November 2010 and September 2021, 1333 studies using EQ-5D-Y were registered by the EuroQol Group.

The primary objective of this study is to summarize published evidence on the psychometric properties of the EQ-5D-Y-3L and 5-level version of EQ-5D-Y (EQ-5D-Y-5L) questionnaire within the framework of a systematic review. The secondary goal is to identify knowledge gaps, which should be filled by future validation studies.

Methods

We followed the Preferred Reporting Items for Systematic Reviews and Meta-Analyses 2020 guidance on systematic reviews and meta-analyses.⁹ The protocol of the study was registered in the PROSPERO database before the start of the project (CRD42021252757).

Search Strategy

We performed a computerized search of the published literature using MEDLINE (through PubMed), Embase (through Elsevier access), and the database of EQ-5D references on the EuroQol website (<https://euroqol.org/search-for-eq-5d-publications/>). The search strategy included a combination of free-text terms and Medical Subject Headings related to the EQ-5D-Y questionnaire. No time limits were used. The details of the searches may be found in [Appendix Tables S1 and S2](#) in Supplemental Materials found at <https://doi.org/10.1016/j.jval.2022.05.013>. Reference lists of identified original articles and reviews were also hand searched to identify additional articles that report using the EQ-5D-Y questionnaire.

Inclusion Criteria

We included original studies published in full text in peer-reviewed journals. Articles had to report EQ-5D-Y (EQ-5D-Y-3L or EQ-5D-Y-5L) measurement or psychometric properties assessed in children, adolescents, or adult respondents acting as a proxy for children. We excluded studies using prefinal EQ-5D-Y-3L versions, reviews, conference articles, and studies not available in full text or published in languages other than English. The detailed inclusion and exclusion criteria are summarized in [Appendix Table S3](#) in Supplemental Materials found at <https://doi.org/10.1016/j.jval.2022.05.013>.

Study Selection

EndNote X7 (Thomson Reuters) software was used to gather records from all databases, remove duplicates, and identify and remove conference abstracts and publications in languages other than English. Titles and abstracts of remaining records were independently screened by 2 authors using the prespecified eligibility criteria. In the next step, all potentially relevant articles were independently examined in full text, also by 2 authors, to determine whether they meet the inclusion criteria. All discrepancies between the reviewers were resolved by consensus.

Data Extraction

To facilitate the structured review, we used a predetermined data extraction form in Microsoft Excel. Data extraction was

performed by one reviewer (D.G.). Second reviewer (K.M.) revised the extracted data from 20% of included studies. We extracted general information on a study—title, year of publication, study design, country, setting, sample characteristics, patient's population, outcome measures used, mode of questionnaire administration (patient or proxy), whether a proxy was used, a type of proxy, and proxy version of the questionnaire. Further extracted information included data on EQ-5D-Y feasibility (time to complete, the number and proportion of missing values within dimensions), distributional properties (the number and percentage of best [ceiling] and worst [floor] level of health in each dimension and across all dimensions ["11111" and "33333"/"55555" health states, respectively]), and the main psychometric properties (reliability, validity, and responsiveness).

We extracted information on the 3 different forms of reliability: test-retest reliability (whether the instrument is able to reproduce the same value on 2 separate administrations when there was no change in health over time), interrater reliability (when administered to different raters, such as self-reported and reported by proxy), and intermodal reliability (when administered by different methods, ie, paper-and-pencil vs web-based versions). We did not include hypothetical data on internal consistency because it is not a relevant psychometric property for the EQ-5D-Y questionnaire. Agreement between 2 applications of the measure is usually assessed with intraclass correlation coefficient (ICC) or Pearson's correlation coefficient for continuous variables (such as EQ VAS or the EQ-5D-Y index) and with Cohen's kappa, percentage of agreement, or Gwet's agreement coefficient for categorical variables (EQ-5D-Y dimensions). We interpreted ICC and kappa according to the Cicchetti¹⁰ guidelines: <0.40, poor; 0.40 to 0.59, fair; 0.60 to 0.74, good; and 0.75 to 1.00, excellent.

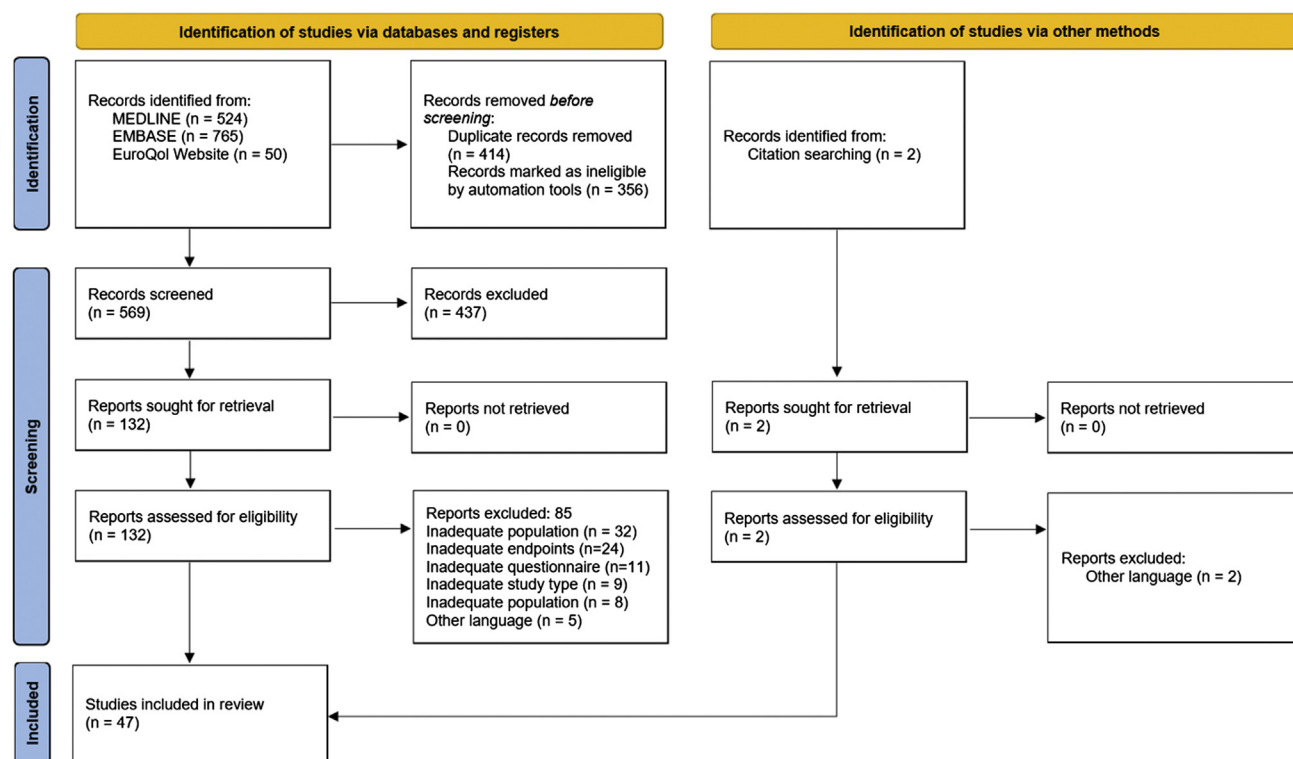
We collected evidence on known-groups validity (the ability to differentiate between a priori determined groups) and convergent validity (the strength of association between the studied instrument and other instruments intended to measure the same construct). We interpreted correlation coefficients as moderate (0.41-0.60) or good (0.61-0.80), according to Landis and Koch criteria.¹¹

We searched for evidence of EQ-5D-Y responsiveness (the ability of the instrument to capture change over time when change is expected). We extracted data on effect size (ES), standardized ES, and standardized response mean.¹² ES, standardized ES, and standardized response mean were interpreted as small (<0.5), moderate (0.5-0.8), or large (>0.8).¹³

Quality Assessment

To assess the risk of bias, we used a modified quality index. The quality index was originally developed and validated by Downs and Black¹⁴ as a 27-item instrument to rate quality of both randomized and nonrandomized studies of healthcare interventions. Given that psychometric studies usually have cross-sectional or longitudinal design, we decided to use its modified version, excluding items related to randomization, blinding, withdrawals, dropouts, and intervention integrity prepared and tested by Ferro et al¹⁵ and used in other systematic reviews of psychometric studies.¹⁶ We furtherly modified this version by removing 3 items related to external validity (representativeness of patients, sample, staff, place, and facilities), but adding one item related to imprecision of the study (size of the sample).

In each study, we seek for a clear description of inclusion/exclusion criteria, hypotheses, outcomes, characteristics of the studied population, main findings, response rate, missing data, and the way authors dealt with it. We also assessed the use of

Figure 1. Literature search and selection process (PRISMA flowchart).

PRISMA indicates Preferred Reporting Items for Systematic Reviews and Meta-Analyses.

actual probability values, evidence of data dredging, appropriateness of statistical tests, validity and reliability of main outcome measures, sample size/power calculation, and study imprecision. Items were scored 0 (no/unable to determine) or 1 (yes). Study imprecision was assessed as 0 (<50 participants), 1 (50–100 participants), or 2 (>100 participants). In total, 13 items give a maximum score of 14. Higher scores indicate studies with higher methodological quality. Quality assessment of included studies was performed by one of the authors (D.G.).

Analysis

Measurement properties were summarized based on the type of property assessed (reliability, validity, and responsiveness) in a tabular and narrative form. The evidence was recapped separately by different health conditions. We tried to explain the heterogeneity in results between studies based on studies characteristics (eg, population, disease, country, comparative instrument used). The following data were pooled using random effects models and inverse variance weights: missing values within the EQ-5D-Y dimensions and EQ VAS, the proportion of “no problems”/“extreme problems” answers within dimensions, the proportion of best possible health state on the EQ-5D-Y profile (“11111”), and correlations (Spearman’s or Pearson’s Rho) with other measures. The negative signs of correlation coefficients were changed if authors had not adjusted for the directionality of the scales. Meta-analysis was performed using the R packages “meta” and “metafor.”^{17,18}

Results

A systematic search of 3 databases run on June 13, 2021, resulted in 1339 references, of which 134 underwent full-text

screening. A total of 87 articles were excluded—the most common reasons for this were inadequate population and inadequate endpoints. Finally, 47 studies were included in the review (see Fig. 1 for the Preferred Reporting Items for Systematic Reviews and Meta-Analyses chart, Appendix Table S4 in Supplemental Materials found at <https://doi.org/10.1016/j.jval.2022.05.013> for the list of excluded studies with reasons, and reference list^{8,19–64} for included studies).

A total of 44 articles included EQ-5D-Y-3L and 7 included EQ-5D-Y-5L. Most studies reported EQ-5D-Y dimensions (89.4%) and EQ VAS (76.6%). Of 12 studies describing the EQ index, 50.0% used the UK MVH A1 (3-level version of EQ-5D based) value set. The country with the highest number of studies was Sweden (n = 10), followed by Spain and United Kingdom (both n = 7) and South Africa (n = 6). Most studies were cross-sectional in design (80.9%) and used the pencil-and-paper version of the questionnaire (74.5%) (Table 1). Individual studies included from 13 to 8919 children and from 13 to 654 proxies (parents or children’s caregivers were used as proxy respondents in 13 studies). The mean age of children was 11.8 years (47.7% males). See Table 2^{8,19–64} for detailed characteristics of the included studies and Appendix Figure S1 in Supplemental Materials found at <https://doi.org/10.1016/j.jval.2022.05.013> for the relationship of the mean child age in the study and the type of respondent.

Quality Assessment

Included studies were of high quality according to the quality index—with mean 11.8 (SD 1.5) points (maximum of the scale, 14 points). Only 3 studies (6.7%) scored below 10 points. The most problematic domains were “sample size or power calculation” and “clear description of hypotheses” (presented in 21.3% and 51.1% of

Table 1. Number of included studies, by category.

Study characteristics	n	%
EQ-5D-Y version		
EQ-5D-Y-3L	40	85.1
EQ-5D-Y-5L	7	6.4
EQ-5D-Y-3L and EQ-5D-Y-5L	4	8.5
Reported EQ-5D-Y outcomes		
EQ-5D-Y dimensions	42	89.4
EQ VAS	36	76.6
EQ index	12	25.5
EQ-5D-Y summary score	4	8.5
Value set		
Australian (EQ-5D-3L)	1	8.3
Chinese (EQ-5D-3L, EQ-5D-5L)	1	8.3
European VAS (EQ-5D-3L)	1	8.3
UK MVH A1 (EQ-5D-3L)	6	50.0
United States (EQ-5D-3L)	1	8.3
United States (EQ-5D-Y, Craig 2016 ⁷⁴)	1	8.3
No data	1	8.3
Country/district		
Australia	1	2.1
Canada	1	2.1
China	3	6.4
Germany	4	8.5
Hong Kong	3	6.4
Italy	2	4.3
Japan	1	2.1
The Netherlands	2	4.3
South Africa	6	12.8
South Korea	2	4.3
Spain	7	14.9
Sweden	10	21.3
Taiwan	1	2.1
Thailand	1	2.1
United Kingdom	7	14.9
Disease group		
General population	14	29.8
Gastrointestinal diseases	1	2.1
Hematological diseases	2	4.3
Immune system disorders	2	4.3
Metabolism and nutrition disorders	4	8.5
Musculoskeletal diseases and orthopedics	10	21.3
Psychiatric disorders	2	4.3
Renal and urinary diseases	1	2.1
Respiratory diseases	2	4.3
Mixed populations	9	19.1
Study design		
Cross-sectional	38	80.9

*Continued in the next column***Table 1.** Continued

Study characteristics	n	%
Prospective cross-sectional	6	12.8
Longitudinal	2	4.3
Qualitative study	1	2.1
Mode of recruitment		
Convenience sample	39	83.0
Random sample	5	10.6
No data	3	6.4
Mode of administration*		
Paper and pencil	35	74.5
Online/electronic	10	21.3
Telephone	1	2.1
No data	6	12.8
Other HRQoL tools in the study†		
EQ-5D-3L	2	4.3
EQ-5D-5L	1	2.1
CHU-9D	3	6.4
HUI	2	4.3
KIDSCREEN-10	5	10.6
KIDSCREEN-27	6	12.8
PedsQL	10	21.3
General health item	8	17.0
SDQ	7	14.9

CHU-9D indicates Child Health Utility 9D; EQ-5D-3L, 3-level version of EQ-5D; EQ-5D-5L, 5-level version of EQ-5D; EQ-5D-Y-3L, 3-level version of EQ-5D-Y; EQ-5D-Y-5L, 5-level version of EQ-5D-Y; HRQoL, health-related quality of life; MVH, Measurement and Value of Health; PedsQL, Pediatric Quality of Life Inventory; SDQ, strength and difficulties questionnaire; VAS, visual analog scale. *Five studies used 2 modes of administration.

†Instruments used at least in 2 studies (except EQ-5D-5L).

articles, respectively; see [Appendix Table S5](#) in Supplemental Materials found at <https://doi.org/10.1016/j.jval.2022.05.013>).

Heterogeneity

The included studies differed significantly in terms of the characteristics of the study population (age, country of origin, general population, or patient population), within the clinical area (different diseases), within conditions (various stages of advancement), and in terms of the quality of life instrument chosen as a comparison for the EQ-5D-Y. Consequently, a substantial statistical heterogeneity ($I^2 \geq 80\%$) was noted in most performed meta-analyses.

Feasibility

The time to complete the EQ-5D-Y (6 studies; $n = 1256$ children) questionnaire ranged from 1.3 to 1.8 minutes in school children,^{25,32,48} while being <5 minutes in children with psychiatric disorders or a mixed population of healthy, ill, and disabled children,^{53,39} and was approximately 8 minutes in children with cerebral palsy.³⁸

The proportion of missing values in the EQ-5D-Y-3L profile, reported in 15 studies, was < 5% in 17 of 18 subpopulations

Table 2. Characteristics of the included studies.

Ref.	Study ID	Tool	Country	Sample size	Study design	Study population/disease area	% male	Age, mean (SD)	Age range
19	Eidt-Koch, 2009	3L	Germany	96	Cross-sectional	Cystic fibrosis	50.0	13.0 (2.5)	8-17
20	Boyle, 2010	3L	England	1771	Cross-sectional	School (pupils)	51.7	13.2 (1.2)	11-15
21	Jelsma, 2010a	3L	South Africa	259	Cross-sectional	School (pupils)	50.0	15.5 (1.3)	13-19
22	Jelsma, 2010b	3L	South Africa	628 dyads	Cross-sectional	Open schools (n = 567), special schools (n = 61)	45.0	nd	7-12
8	Ravens-Sieberer, 2010	3L	5 countries (Germany, Italy, South Africa, Spain, and Sweden)	2809	Cross-sectional	School (pupils)	50.4	13.3	8-19
23	Wu, 2010	3L	Canada	3421	Cross-sectional	School (grade 5 students)	48.6	nd	10-11
24	Scalone, 2011	3L	Italy	440	Cross-sectional	School (pupils, n = 415), acute lymphoblastic leukemia (n = 25)	47.9 (general population), 72% (ALL)	9.4	8-13
25	Canaway, 2013	3L	UK	160	Cross-sectional	School (pupils)	57.0	6.4	6-7
26	Oluboyede, 2013	3L	UK	49	Cross-sectional	School (pupils)	46.9	12.7 (1.4)	11-18
27	Burström, 2014	3L	Sweden	481	Cross-sectional	School (pupils, n = 407), functional disabilities (n = 71; arthrogryposis multiple congenita, myelomeningocele, cerebral palsy, orthopedic lower limb deformities, juvenile idiopathic arthritis, and achondroplasia)	39.4	12.0 (3.1)	7-17
28	Domellöf, 2014	3L	Sweden	175	Cross-sectional	Autism spectrum disorders, intellectual disabilities, movement disorders, hearing disabilities	67.4	11.8	7-17
29	Ortqvist, 2014	3L	Sweden	115	Cross-sectional	Knee injuries	44.3	13.0 (1.9)	7-16
30	Bergfors, 2015	3L	Sweden	94	Cross-sectional	Asthma	57.4	12.1 (2.4)	8-16
31	Chen, 2015b	3L	Australia	2020	Cross-sectional	General population aged 11-17 years	49.3	14 (2)	11-17
32	Robles, 2015	3L	Spain	713	Cross-sectional	School (pupils)	46.0	11.7 (2.8)	7-18
33	Sierwald, 2016	3L	Germany	425	Cross-sectional	School (pupils, n = 313); orthodontic patients (n = 112)	54.1	10.4	7-17
34	Bray, 2017	3L	UK	13 dyads	Cross-sectional	Children with mobility impairments (85% cerebral palsy)	61.5	nd	6-18
35	Herngren, 2017	3L	Sweden	25	Cross-sectional	Slipped capital femoral epiphysis (SCFE)	nd	nd	8-15
36	Kim, 2017	3L	South Korea	2466	Cross-sectional	School (pupils)	51.4	9.53 (1.7)	7-12
37	Murillo, 2017	3L	Spain	136	Cross-sectional	T1DM	41.1	13.45 (2.9)	8-19
38	Perez Sousa, 2017	3L	Spain	62 triples (child, mother, father)	Cross-sectional	Cerebral palsy	56.4	10.09 (2.31)	6-17
39	Scott, 2017	3L	South Africa	224	longitudinal	Mainstream school (n = 105), special school (physical disability) (n = 35), chronically ill (n = 32), acutely ill (n = 52)	nd	10.5 (1.45)	8-12

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Table 2. Continued

Ref.	Study ID	Tool	Country	Sample size	Study design	Study population/disease area	% male	Age, mean (SD)	Age range
40	Åström, 2018	3L	Sweden	6574	Cross-sectional	School (pupils)	50.6	15.9 (1.6)	13-18
41	Hsu, 2018	3L	Taiwan	68	Prospective cross-sectional	Chronic kidney disease	64.7	11.96 (4.08)	7-18
42	Kim, 2018	3L	South Korea	8919	Cross-sectional	Allergy symptoms	51.4	10.2 (1.8)	7-13
43	Perez Sousa, 2018	3L	Spain	151 dyads	Cross-sectional	Overweight and obese children	53.0	nd	6-14
44	van Summeren, 2018	3L	The Netherlands	56 dyads	Cross-sectional	Functional constipation	42.9	Median 10.0 (IQR 8.3-12.0)	8-18
45	Löf, 2019	3L	Sweden	215	Cross-sectional	Idiopathic clubfoot with or without neurodevelopmental difficulties (n = 106), school (pupils, n = 109)	73.0	9.4 (0.6)	8-10
46	López-Bastida, 2019	3L	Spain	267 dyads	Cross-sectional	T1DM	53.2	11.0 (3.9)	2-18
47	Mayoral, 2019	3L	Spain	136	Longitudinal	T1DM	47.1	13.99 ± 0.2	8-19
48	Scott, 2019	3L	South Africa	64	Cross-sectional	Juvenile idiopathic arthritis	37.5	11.8 (2.53)	8-15
49	Shiroyiwa, 2019	3L	Japan	654 dyads	Prospective cross-sectional	General population	50.0	11.5 (2.3)	8-15
50	Wong, 2019a	3L/5L	Hong Kong, China	129	Prospective cross-sectional	Idiopathic scoliosis	19.4	14.0 (1.9)	8-17
51	Wong, 2019b	3L/5L	Hong Kong, China	129	Prospective cross-sectional	Idiopathic scoliosis	19.4	14.0 (1.9)	8-17
52	Acaster, 2020	3L	UK	100 dyads	Cross-sectional	Peanut allergy	58.0	9.82 (3.42)	4-15
53	Åström, 2020	5L	Sweden	52	Cross-sectional	Psychiatric disorders	17.0	15.4 (1.4)	13-17
54	Lin, 2020	3L	Hong Kong, China	130 dyads	Cross-sectional	Idiopathic scoliosis	9.5	Median 14.0	11-16
55	Pan, 2020	3L	China	3842	Cross-sectional	School (pupils)	52.1	12.1	7-18
56	Ryan, 2020	3L	England	63	Cross-sectional	Cerebral palsy	58.7	13.7 (2.5)	10-19
57	Sinlapamongkolkul, 2020	3L	Thailand	150 dyads	Cross-sectional	Thalassemia	54.0	9.1 (4.3)	2-18
58	Verstraete, 2020b	3L	South Africa	321 proxies	Cross-sectional	Caregivers of acutely ill (40%), chronically ill (31%), general population (29%) children	53.0	4.0	3-5
59	Wright, 2020	3L	England	6328	Cross-sectional	School (pupils)	46.4	9.0	12-16
60	Doeleman, 2021	5L	The Netherlands	68	Cross-sectional	Juvenile idiopathic arthritis	31.0	Median 13.6 (IQR 10.5-16.4)	nd
61	Krig, 2021	5L	Sweden	52	Qualitative study	Psychiatric disorders	17.0	15.4	13-17

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Table 2. Continued

Ref.	Study ID	Tool	Country	Sample size	Study design	Study population/disease area	% male	Age, mean (SD)	Age range
62	Ludwig, 2021	3L	Germany	223	Prospective cross-sectional	T1DM (63%), rheumatic disorders (37%)	42.0	11.9	8-15
63	Pei, 2021	3L/5L	China	262	Cross-sectional	School (pupils)	41.6	12.7 (3.1)	8-18
64	Zhou, 2021	3L/5L	China	96 dyads	Prospective cross-sectional	Hematological malignancies (47.9% acute lymphoblastic leukemia)	64.6	10.5 (2.2)	8-17

3L indicates 3-level version of EQ-5D; 5L, 5-level version of EQ-5D; ALL, acute lymphoblastic leukemia; ID, identification; IQR, interquartile range; nd, no data; Ref., reference; T1DM, type 1 diabetes mellitus; UK, United Kingdom.

(94.4%). Similarly, the proportion of missing values in the EQ-5D-Y-5L profile was <5% in 4 of 4 reporting studies (100%). The pooled proportion of missing profiles was 0% (95% confidence interval [CI] 0-2; $I^2 = 85\%$) and 1% (95% CI 0-2; $I^2 = 0\%$) for EQ-5D-Y-3L and EQ-5D-Y-5L, respectively (see Fig. 2 and Appendix Fig. S2 in Supplemental Materials found at <https://doi.org/10.1016/j.jval.2022.05.013>). Individual EQ-5D-Y dimensions had a pooled 0.4% of missing values (except from having pain or discomfort and feeling worried, sad, or unhappy dimensions, with 0.5% missing values; see Appendix Table S6 in Supplemental Materials found at <https://doi.org/10.1016/j.jval.2022.05.013>). The proportion of missing EQ VAS scores, reported in 19 studies, was acceptable in 19 of 23 subpopulations (82.6%). The pooled value amounted to 1% (95% CI 1-2; $I^2 = 98\%$; see Appendix Fig. S3 in Supplemental Materials found at <https://doi.org/10.1016/j.jval.2022.05.013>). The highest value (13.8%) was reported in a general population study from South Korea.⁴²

Distribution Properties

The pooled proportion of respondents reporting the best health ("11111") in EQ-5D-Y-3L (so called "ceiling effect"), based on the data from 16 studies (23 subpopulations), was 55% (95% CI 46-63; $I^2 = 99\%$) and ranged from 23% (95% CI 18-29; $I^2 = 0\%$) in hematological diseases to 64% (95% CI 55-72; $I^2 = 100\%$) in the general population (Fig. 3). Similarly, the pooled proportion of "11111" in EQ-5D-Y-5L, based on the data from 4 studies (5 subpopulations), was 20% (95% CI 5-53; $I^2 = 91\%$). It ranged in individual studies from 0% (95% CI 0-7) in patients with psychiatric problems to 59% (95% CI 50-67) in children with idiopathic scoliosis (see Appendix Fig. S4 in Supplemental Materials found at <https://doi.org/10.1016/j.jval.2022.05.013>). The most severe state in EQ-5D-Y-3L ("33333") was reported in 5 studies (for 7 subpopulations). The pooled value amounted to 0.04% (95% CI 0.02-0.10; $I^2 = 0\%$; see Appendix Table S7 in Supplemental Materials found at <https://doi.org/10.1016/j.jval.2022.05.013>). There were no data on the most severe EQ-5D-Y-5L state ("55555") in studies included in the review.

Reliability

Ten articles addressed test-retest reliability in 21 subpopulations, with the mean interval time ranging from 12 hours to 9 months (see Appendix Table S8 in Supplemental Materials found at <https://doi.org/10.1016/j.jval.2022.05.013>). The EQ VAS test-retest reliability (ICC) was excellent in 8 of 12 subpopulations (67%), good in 2 (16.7%), fair in 1 (8.3%), and poor in 1 subpopulation, where proxy respondents were used.⁴⁹ Agreement

between 2 measurements of the EQ-5D-Y summary score was reported in one study⁴⁷ and found to be excellent. The ICC of EQ-5D-Y index measurements gave mixed results (from fair to excellent) in 1 study, where proxy respondents for 3- to 5-year-old children were used.⁵⁸ Test-retest reliability results for individual EQ-5D-Y dimensions gave mixed results when using kappa or weighted kappa—poor to excellent for mobility (MO) (0.22-0.79), self-care (SC) (0.17-0.95), and usual activities (UA) (0.20-0.78) and poor to fair for pain/discomfort (PD) (0.01-0.45) and anxiety/depression (AD) (−0.15-0.55). The percentage of agreement was >70% in 80% of 78 measurements and >80% in 57% cases. It tended to be lower for the PD and AD dimensions.

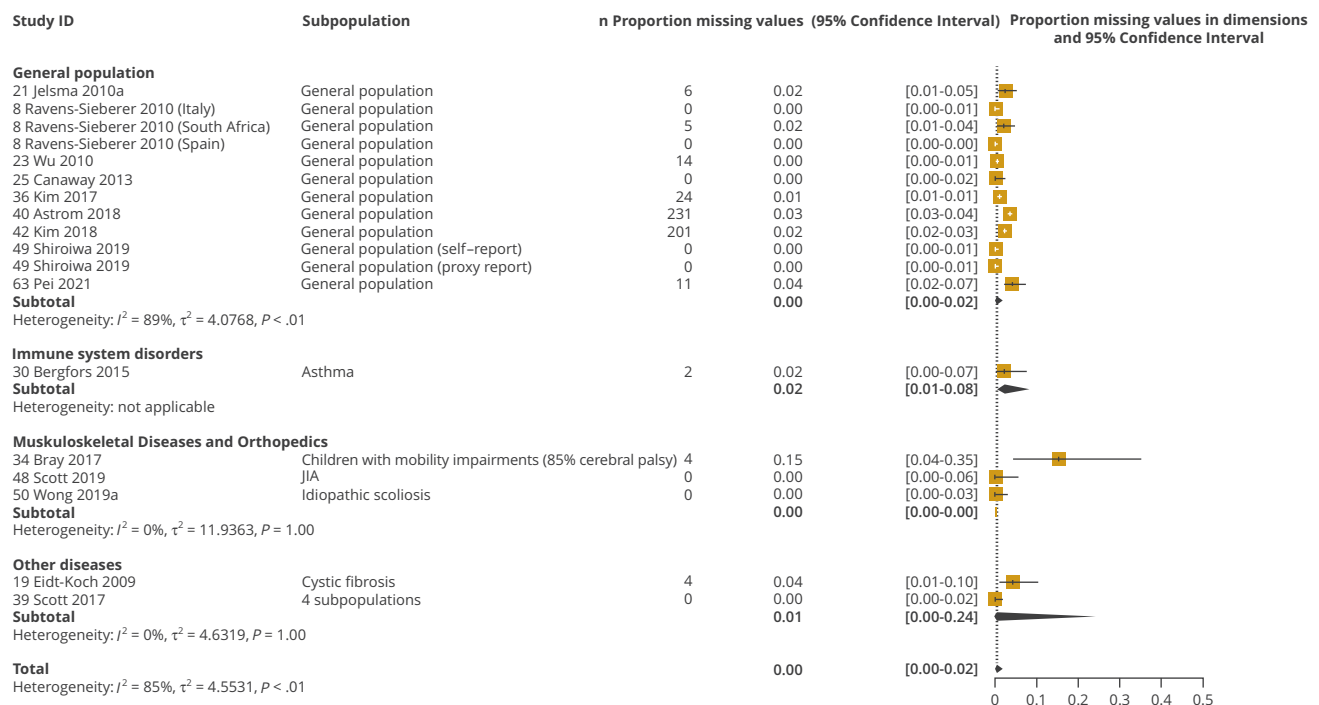
Ten studies investigated interrater reliability with EQ-5D-Y administered to both children and their parents. Three studies used "proxy 1," another 3 used "proxy 2," and the undefined version of EQ-5D-Y was used in 4 studies. Interrater agreement, measured with kappa in individual studies, tended to be higher in MO (0.04-0.71) than other dimensions (SC 0.04-0.36, UA 0.01-0.44, PD 0.07-0.41, AD 0.02-0.42). The details of available evidence on the agreement between EQ-5D-Y self-report and proxy report are presented in Appendix Table S9 in Supplemental Materials found at <https://doi.org/10.1016/j.jval.2022.05.013>.

Intermodal reliability was addressed in one study of 713 pupils filling both the paper-and-pencil and web-based versions of the EQ-5D-Y-3L in random order with a mean interval of at least 2 hours.³² The authors found an excellent agreement for EQ VAS and MO; good agreement for SC, PD, and AD; and fair for UA.

Validity

The EQ-5D-Y known-groups validity was assessed in 27 studies, including 18 investigating EQ-5D-Y dimensions, 17 using EQ VAS, and 7 using the EQ-5D-Y index. Only 9 articles presented formal hypotheses on the direction of group differences. In general, the EQ-5D-Y can distinguish across age groups, lower and higher grades, boys and girls, children from the general population and patients (children with allergy, shortsightedness, idiopathic clubfoot, acute lymphoblastic leukemia, etc), and disease severity (eg, in juvenile arthritis). The details of the known-groups validity analyses are presented in Appendix Table S10 in Supplemental Materials found at <https://doi.org/10.1016/j.jval.2022.05.013>.

The convergent validity of EQ-5D-Y was investigated in 21 articles. Concerning the EQ-5D-Y index, the strongest correlation was found with other global measures of health (most results for the Pediatric Quality of Life Inventory; correlation coefficient = 0.49; 95% CI 0.41-0.56; $I^2 = 52\%$). The weakest were those with clinical and biological measures (0.21; 95% CI 0.12-0.30; $I^2 = 44\%$) and measures of school functioning (0.29; 95% CI 0.25-0.33; I^2 not

Figure 2. Proportion of missing values in EQ-5D-Y-3L profile pooled across health conditions.

EQ-5D-Y-3L indicates 3-level version of EQ-5D-Y; ID, identification; JIA, juvenile idiopathic arthritis.

applicable) (Fig. 4). The strength of association of EQ VAS was high with activities of daily living (ADLs) instruments (0.56; 95% CI 0.41-0.68; $I^2 = 0\%$), multiattribute utility instruments (0.46; 95% CI 0.28-0.61; $I^2 = 72\%$), and other global measures of health (0.40; 95% CI 0.44-0.48; $I^2 = 75\%$). The poorest correlation was once again observed with school functioning (0.27; 95% CI 0.19-0.35; $I^2 = 33\%$; see Appendix Fig. S5 in Supplemental Materials found at <https://doi.org/10.1016/j.jval.2022.05.013>). Within the EQ-5D-Y dimensions, the strongest correlation was observed for the UA and ADL instruments (0.54; 95% CI 0.38-0.67), followed by PD and ADL (0.46; 95% CI 0.17-0.68; see Appendix Table S11 in Supplemental Materials found at <https://doi.org/10.1016/j.jval.2022.05.013>).

An article by Krig et al⁵¹ used qualitative methods to assess the content validity of the EQ-5D-Y-5L questionnaire in children and adolescents within psychiatric inpatient care. The study results indicated that these patients, in general, could self-report their health in a meaningful way with the EQ-5D-Y-5L questionnaire. The inclusion of physical health dimensions and the generic character of the instrument were regarded positively. Nevertheless, some patients considered the EQ-5D-5L descriptive system to be too generic.

Responsiveness

The ability to capture changes in health over time by EQ-5D-Y-3L and EQ-5D-Y-5L was investigated in 4 studies and a single study, respectively (see Appendix Table S12 in Supplemental Materials found at <https://doi.org/10.1016/j.jval.2022.05.013>). The responsiveness of EQ VAS was addressed in 3 studies,^{39,43,47} whereas that of the EQ-5D-Y index was assessed in 2 studies^{39,51} and the EQ-5D-Y summary score in a single study.⁴⁷ A study of type 1 diabetes mellitus children who improved over an assessment period of 9 months showed moderate (ES ≥ 0.7)

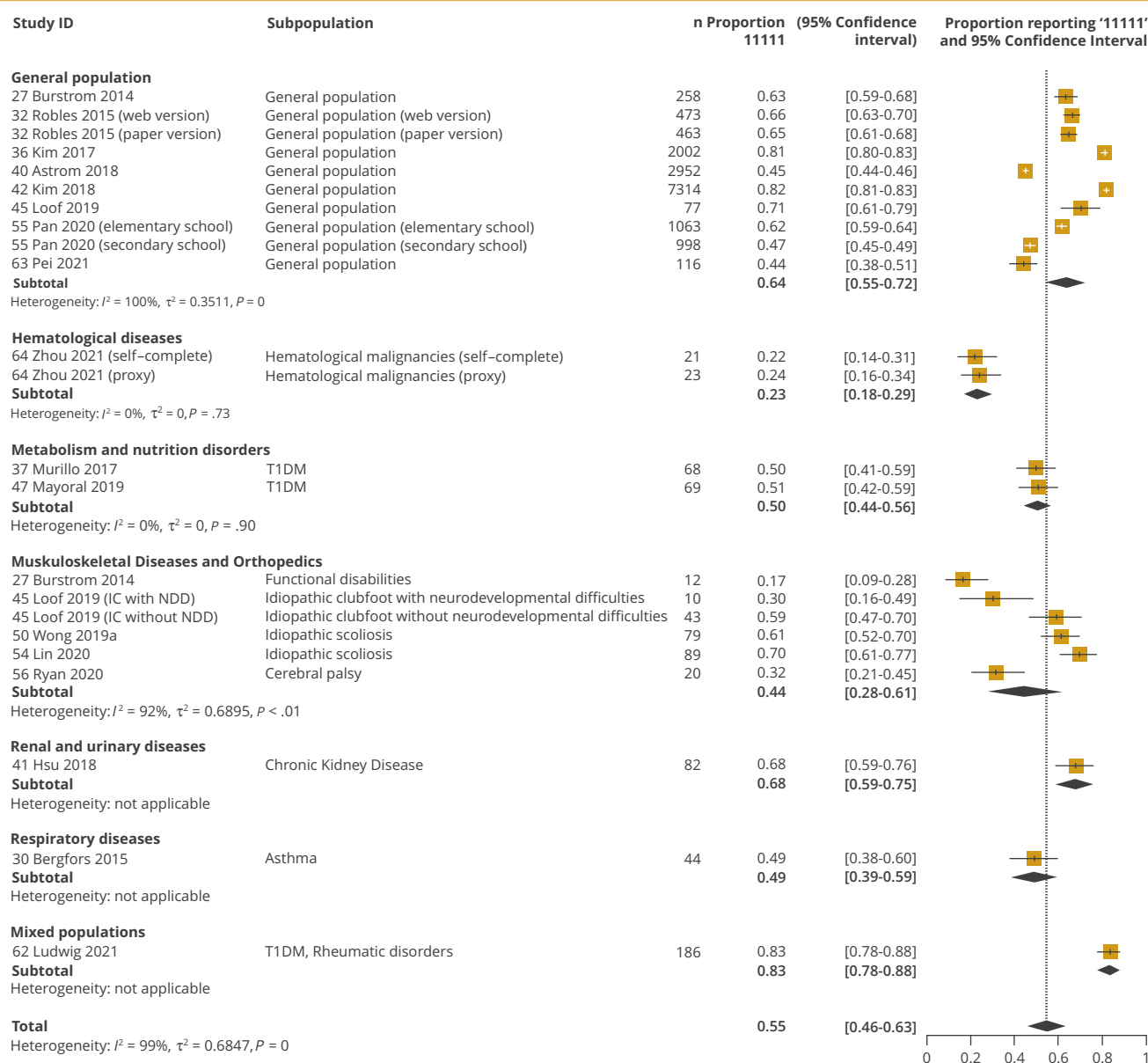
responsiveness for EQ VAS and the EQ summary score.⁴⁷ A article on children with idiopathic scoliosis who deteriorated over 3 months found at least moderate responsiveness for the EQ-5D-Y-3L and EQ-5D-Y-5L indices, based on 75% of calculated statistics.⁵¹

Discussion

We performed a systematic review of the psychometric properties EQ-5D-Y, analyzing 47 studies involving >45 000 children from 15 countries published during the last decade. The review summarizes the available evidence on EQ-5D-Y feasibility, distributional properties, test-retest, interrater and intermodal reliability, known-groups and convergent validity, and responsiveness.

The EQ-5D-Y-3L and EQ-5D-Y-5L profiles exhibit high feasibility. Approximately 95% and 100% of studies, respectively, have a proportion of missing values in the profile of <5%. More significant proportions of missing values were noticed for EQ VAS, which is consistent with observations for the adult version of the questionnaire.⁶⁵

In our review, a substantial ceiling effect (proportion of respondents reporting the best health profile) was noted, especially in general populations of school pupils, but in some patient populations. This psychometric property is expected in the instrument constructed to measure deviations in health, rather than positive health, but should be taken into account during the choice of the study instrument. Introduction of the 5-level version-EQ-5D-Y reduced this ceiling effect in comparison with EQ-5D-Y-3L, which was confirmed by a decrease in the pooled proportion of “11111” states in our review, and in a direct comparison of questionnaires in a study by Wong et al.⁵⁰ There was no floor effect for the EQ-5D-Y-3L (high proportion of “33333” states).

Figure 3. Proportion reporting EQ-5D-Y-3L “11111” profile (“no problems”) pooled across health conditions.

EQ-5D-Y-3L indicates 3-level version of EQ-5D-Y; IC, Idiopathic clubfoot; ID, identification; NDD, neurodevelopmental difficulty; T1DM, type 1 diabetes mellitus.

No study with the use of EQ-5D-Y-5L reported a proportion of “55555” states.

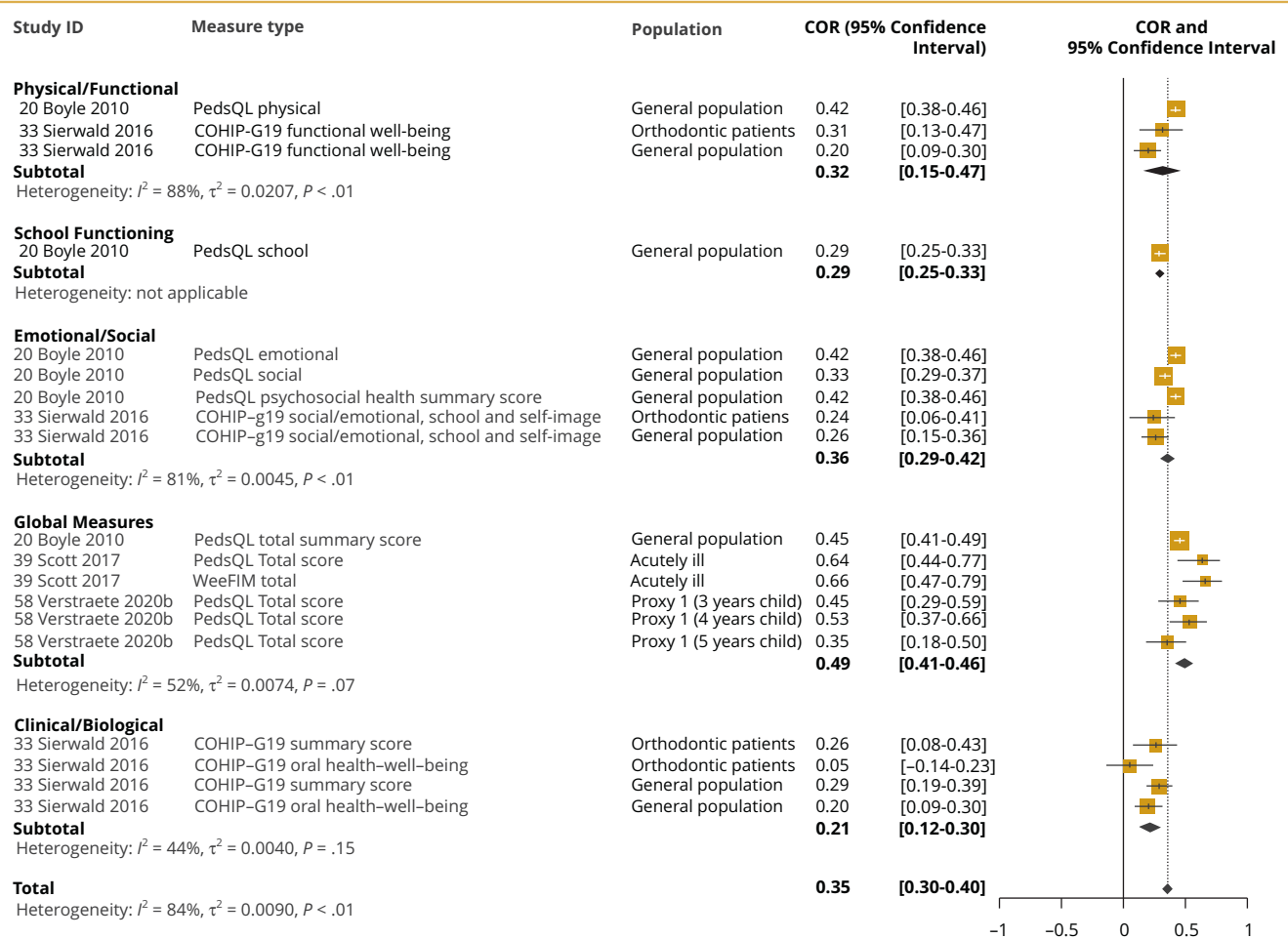
We found 10 articles assessing EQ-5D-Y test-retest reliability in >20 subpopulations. Most of the studies presented results addressing EQ VAS or EQ-5D-Y dimensions. Although EQ VAS test-retest reliability was excellent in two-thirds of populations, mixed results were noted for individual dimensions, especially for PD and AD. This may be due to an excessive interval time between measurements (up to 9 months). Given that the EQ-5D-Y recall period is only 1 day (“today”), the shorter interval times (ie, 12 hours) seem to be appropriate for the EQ-5D-Y test-retest reliability assessment to increase the likelihood of an unchanged health status. The test-retest reliability of the EQ-5D-Y index, estimated based on value sets explicitly constructed for this questionnaire, should hopefully be addressed by studies in the near future. Similarly, the mixed results on EQ-5D-Y interrater reliability should stimulate research that will deepen this thread.

Currently, such results may serve to caution researchers against unnecessary abuse of proxy reporting by parents.

Nearly 30 studies addressed EQ-5D-Y known-groups validity in a wide variety of populations (both general and clinical), countries, and age groups. Unfortunately, only one-third of them presented previous formal hypotheses on the direction of group differences. Convergent validity studies showed that, where the assessment of the child’s functioning at school is required, the EQ-5D-Y, being weakly correlated with this dimension, should be supplemented with other school-specific measures.

Available evidence on EQ-5D-Y responsiveness is significantly limited and needs extensive research. The studies should focus on the responsiveness of the EQ-5D-Y index, calculated based on recently published child-specific value sets.

Our review possesses several limitations. We included articles published only in English. Consequently, some psychometric studies published in Spanish^{66,67} or German⁶⁸ were excluded. We

Figure 4. Pooled correlation coefficient for EQ-5D-Y index with other measures of health.

COHIP indicates Child Oral Health Impact Profile; COR, correlation coefficient; ID, identification; PedsQL, Pediatric Quality of Life Inventory; WeeFIM, Functional Independence Measure for Children.

did not include studies on feasibility or distributional properties only. The study to be included had to report at least one endpoint on reliability, validity, or responsiveness. Addressing the quality of included studies, the most problematic quality index domain was “sample size calculation”—one-fifth of the studies presented such estimations. Nevertheless, most studies involve a large or at least medium population size. Only 30% of the included articles were based on samples <100 children, which is much less than in studies of Health Utilities Index Mark 2 (HUI2) or Health Utilities Index Mark 3 (HUI3) (58% and 42%, respectively).⁶⁹

At least 10 preference-based health measures for use in child and adolescent populations exist (Adolescent Health Utility Measure, Assessment of Quality of Life - 6D, Child Health Utility - 9D, EQ-5D-Y-3L, EQ-5D-5L, HUI2, HUI3, Quality of Well-Being Scale, Sixteen-Dimensional health-related measure, and Seventeen-Dimensional health-related measure). Several reviews describe these instruments in the context of valuation and health state utility values.⁶⁹⁻⁷² Only 1 review comprehensively addressed the psychometric performance of 4 selected measures from this group.⁷³ Based on a systematic search up to March 2019, Rowen et al found 20 studies on the psychometric characteristics of EQ-5D-Y-3L. The authors claimed that although EQ-5D-Y-3L was assessed in fewer studies than HUI3 or HUI2, the proportion of studies that found evidence of good psychometric properties was

more significant. They cautioned that most evidence related to EQ-5D-Y-3L was based on dimensions. Finally, they noticed that none of the 4 assessed instruments (Child Health Utility - 9D, EQ-5D-Y-3L, HUI2, and HUI3) performed strongly for interrater reliability between child self-report and parent proxy report.

Our review, being the second one to assess the psychometric performance of EQ-5D-Y-3L, covered a publication period that was >2 years longer, included double the number of publications, and also addressed the psychometrics of the EQ-5D-Y-5L questionnaire. Based on our findings, we share the view of Rowen et al that future EQ-5D-Y psychometric studies should fully address interrater reliability. Moreover, test-retest reliability and the responsiveness of the EQ-5D-Y index should be researched. In our review, only 12 (26%) of 47 studies reported EQ index calculations. All estimations were based on adult health state utility value sets (half of them on the UK TTO value set). None used a value set devoted to EQ-5D-Y. This is because the official EQ-5D-Y valuation protocol⁵ and the first child-specific value sets^{6,7} were only published recently. Given that Japan and Slovenia have completed their valuation studies and more than a dozen further valuation studies are underway and are expected to finish in 2022, future research estimating the EQ-5D-Y index should rely on child-specific utilities, and psychometric studies should address the performance of the child-specific EQ index.

Rowen et al⁶⁹ found only one study on EQ-5D-Y-5L, which they excluded from their psychometric review. After 2 years, we were able to identify 6 additional articles on this topic.^{51,53,60,61,63,64} Despite this, there are only 2 studies on the psychometric evidence concerning EQ-5D-Y-5L reliability, 3 on validity, and a single study on responsiveness. In consequence, further research on this topic is urgently needed.

Future research should also focus on clinical areas that are either not covered or poorly covered. Although we found a substantial number of studies in the general population (school children) and subpopulations with musculoskeletal, neurological, or orthopedic diseases, there are still clinical areas with no studies whatsoever, such as attention deficit hyperactivity disorder, epilepsy, or cancer (other than hematological malignancies), or single studies, such as asthma, autism, or obesity.

Furthermore, most of psychometric EQ-5D-Y research originates from Europe and Asia. We have not identified any studies from South America, whereas there are only individual studies from countries in Africa and North America.

Conclusions

This systematic review provides a summary of the measurement properties and psychometric performance of EQ-5D-Y-3L and EQ-5D-Y-5L. The existing evidence supports the use of the EQ-5D-Y descriptive system and EQ VAS in children and adolescent populations. Further research on the reliability and responsiveness of the EQ-5D-Y index obtained based on child-specific value sets is needed.

Supplemental Materials

Supplementary data associated with this article can be found in the online version at <https://doi.org/10.1016/j.jval.2022.05.013>.

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